

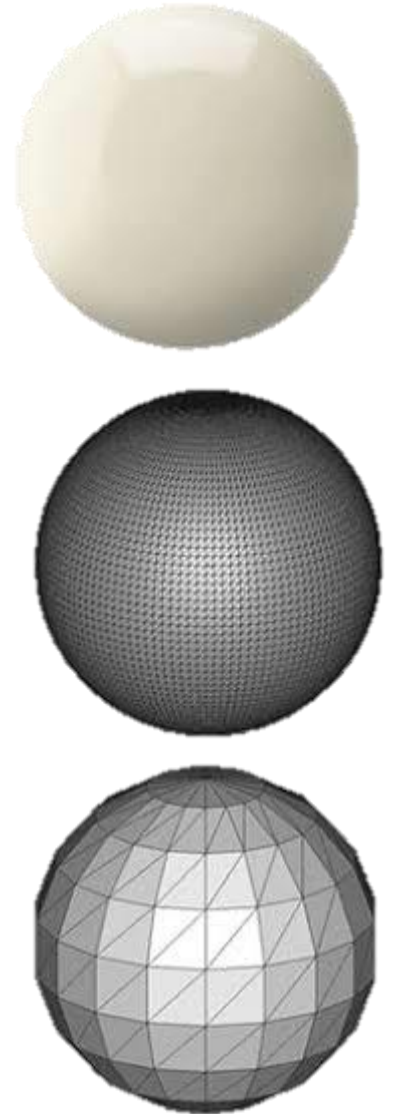
STL file



STL FILE FORMAT



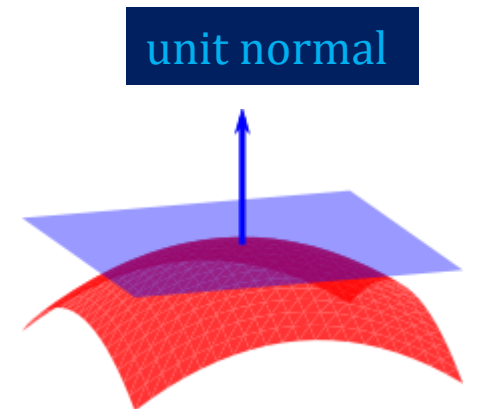
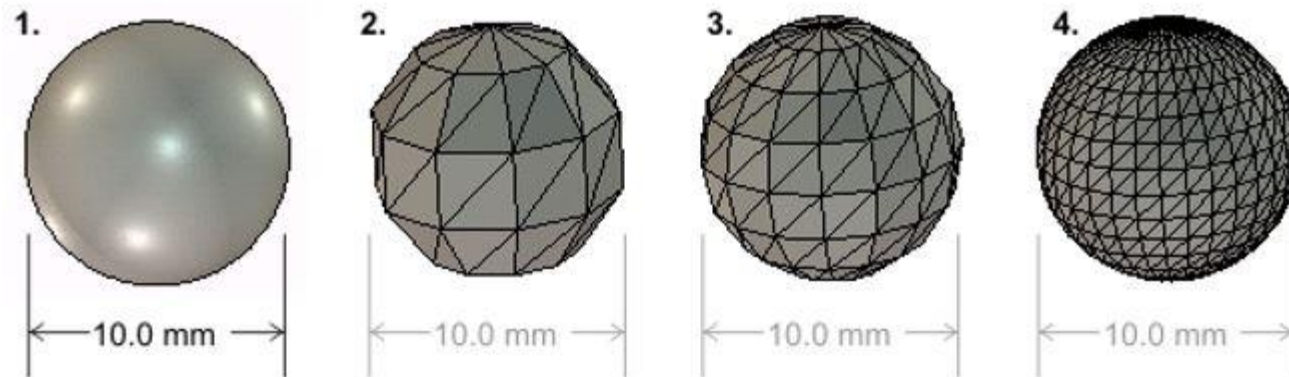
- STL (**Standard Tessellation Language**) is a file format native to the **stereolithography** CAD software created by 3D Systems.
- File format is supported by many other software packages.
- It is widely used for rapid prototyping and computer-aided manufacturing.
- STL files describe only the **surface geometry** of a three-dimensional object **without any representation of colour, texture or other common CAD model attributes**.
- The STL format specifies both **ASCII** and **binary** representations.
- Binary files are **more common**, since they are more compact.



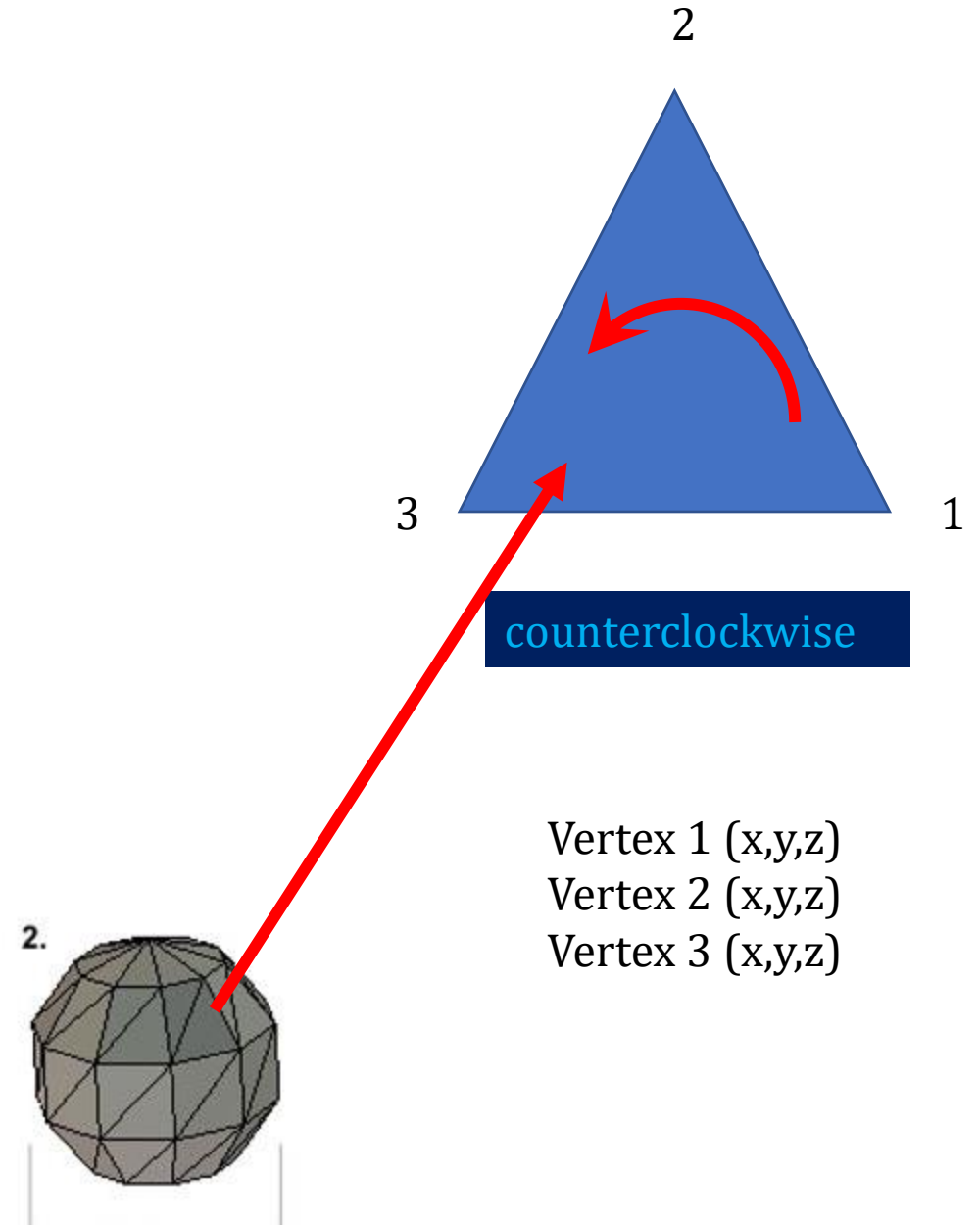
- An STL file **describes** a raw unstructured triangulated surface by the **unit normal** and **vertices of the triangles** using a three- dimensional Cartesian coordinate system.
- Unit normal and vertices of the triangles ordered by the right-hand rule.
- This format is used as industry standard in rapid prototyping.

PROCESS OF APPROXIMATING SURFACES WITH TRIANGLES:

- Each 3D form is made out of polygons.
- A polygon is defined as a flat shape which is bounded by a closed circuit.
- Each polygon with n sides can be represented using $n-2$ triangles

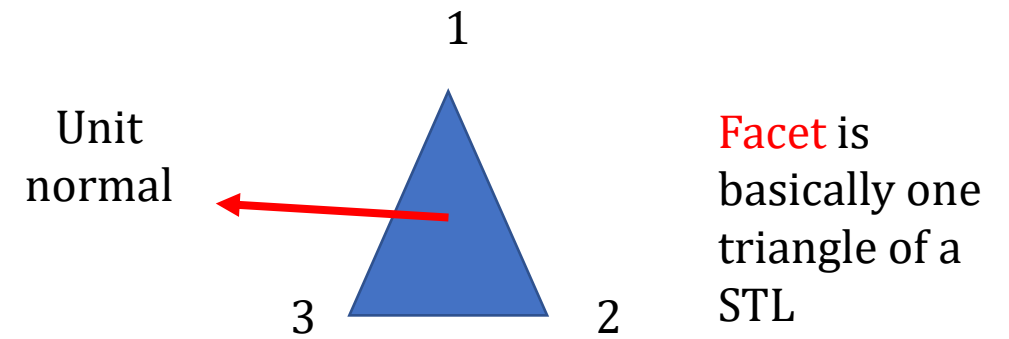


- The outside of the triangle is best defined using a **right-hand** rule approach.
- As we look at a triangle from the outside, the vertices should be listed in a **counterclockwise** order.
- Using the right hand with the thumb pointing upwards, the other fingers curl in the direction of the order of the vertices, the starting vertex being arbitrary

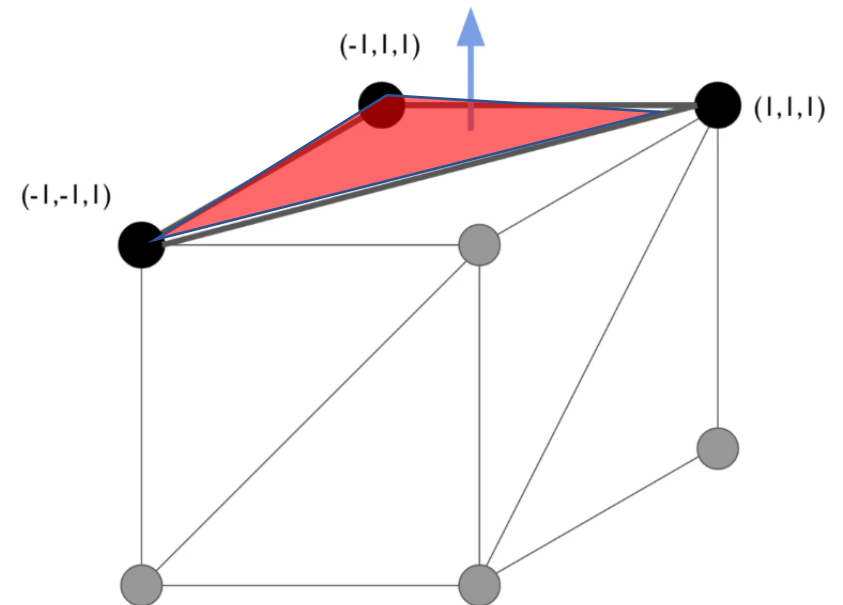


How an STL file looks inside a text editor

```
16 facet normal 0.000000 0.000000 -1.000000
17 outer loop
18 vertex 1.000000 1.000000 -1.000000
19 vertex 1.000000 -1.000000 -1.000000
20 vertex -1.000000 1.000000 -1.000000
21 endloop
22 endfacet
23 facet normal 1.000000 0.000000 0.000000
24 outer loop
25 vertex 1.000000 1.000000 -1.000000
26 vertex 1.000000 1.000000 1.000000
27 vertex 1.000000 -1.000000 1.000000
28 endloop
29 endfacet
30 facet normal 1.000000 0.000000 0.000000
31 outer loop
32 vertex 1.000000 -1.000000 1.000000
33 vertex 1.000000 -1.000000 -1.000000
34 vertex 1.000000 1.000000 -1.000000
```

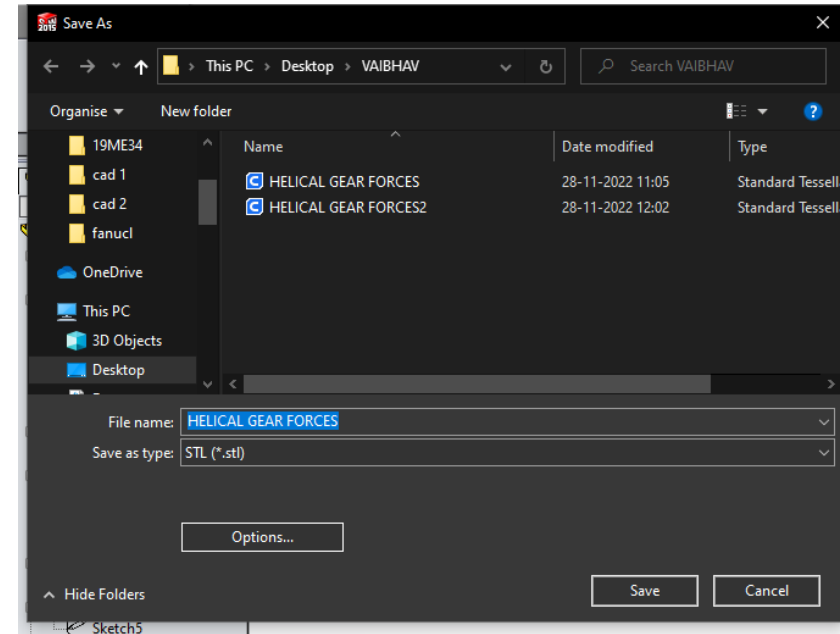


How one facet is represented



Why It is used in RP?

- The STL file format has become the **standard data transmission** format, and is the format required to interact with Quickparts.
- STL format **approximates** the surfaces of a solid model with triangles.
- Almost all of today's CAD systems are capable of producing an STL file.
- For the user the process is often as simple as selecting **File, Save As, and STL**.
- In all cases, export your STL file as a Binary file. This saves on time and file size.



There is absolutely no loss of data whatsoever due to compression and encryption involved in the RP file format. If you create a RP file from a STL file and then convert it back to a STL file, you will end up with exactly the same STL file.

How to Create Stereolithography .stl files (CAD to STL file Conversion)

- A STL is a triangulated representation of a 3D CAD model.
- The triangulation of a surface will cause faceting of the 3D model.
- The parameters used for a STL will affect how much faceting occurs.
- You cannot build the model any better or smoother than the STL file,
- so if the STL is coarse and faceted, that is what you can expect in the final model.

Saving of stl file in solidworks

- In the CAD package, when exporting to STL, you may see parameters for **chord height**, **deviation**, **angle tolerance**, or something similar.
- These are the parameters that affect the faceting of the STL.
- You don't necessarily want to go too small.
- The **finer the STL** the larger the file is in size, which will affect **processing time** in Insight as well as **build time**.

Export Options

File Format

- IGES 5.3
- STEP
- ACIS
- Parasolid
- VRML
- IFC
- STL/AMF
- VDA
- TIF/PSD/JPG/PNG
- EDRW/EPRT/EASM
- PDF

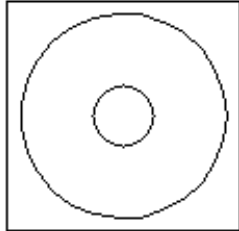
Output as

☒ Binary ☐ ASCII

Unit: **Millimeters**

Resolution

☐ Coarse ☒ Fine ☐ Custom



Deviation

Tolerance: 0.01589042mm

Angle

Tolerance: 10.00000deg

☒ Show STL info before file saving

☐ Preview

Triangles: File size:

☐ Do not translate STL output data to positive space

☐ Save all components of an assembly in a single file

☐ Check for interferences

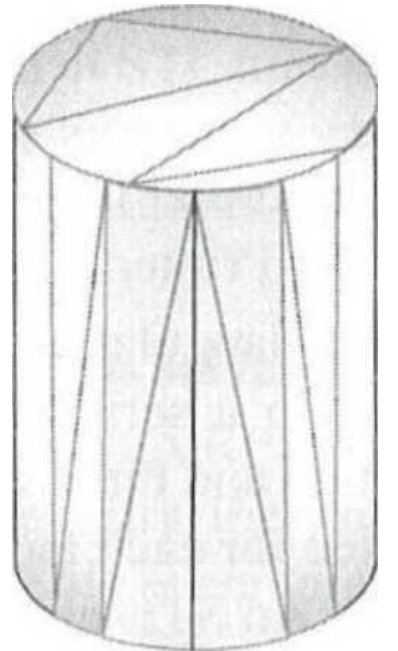
How to Create Stereolithography .stl files (CAD to STL file Conversion)

Part preparation:

- The Rapid Prototyping Process uses .stl or Stereolithography files to build physical 3D Cad Models.
- .Stl files are created using a mesh made from triangles to represent the physical part.
- Most CAD programs allow an easy export of .stl files.

Resolution Too Low:

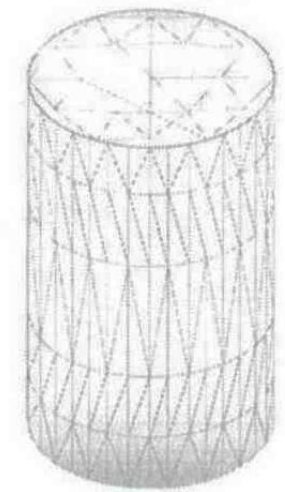
- When the faceting is too coarse you can see flat spots on curved surfaces. The **flat spots** in the .stl file will show up when the part is produced.



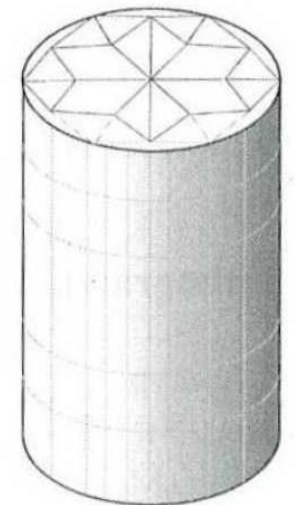
How to Create Stereolithography .stl files (CAD to STL file Conversion)

Resolution Too High

- While the parts will print when the resolution is too high, it can cause **delays in processing** parts because of large size. **Increasing the resolution excessively does not improve the quality of the produced parts.**



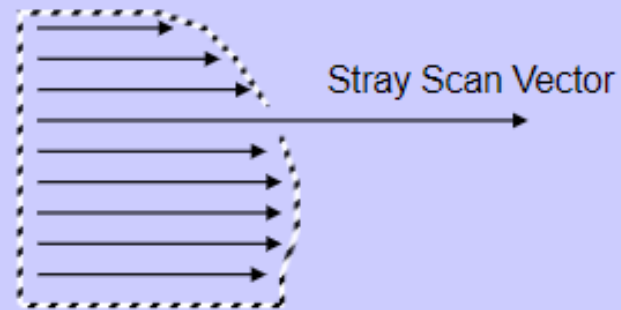
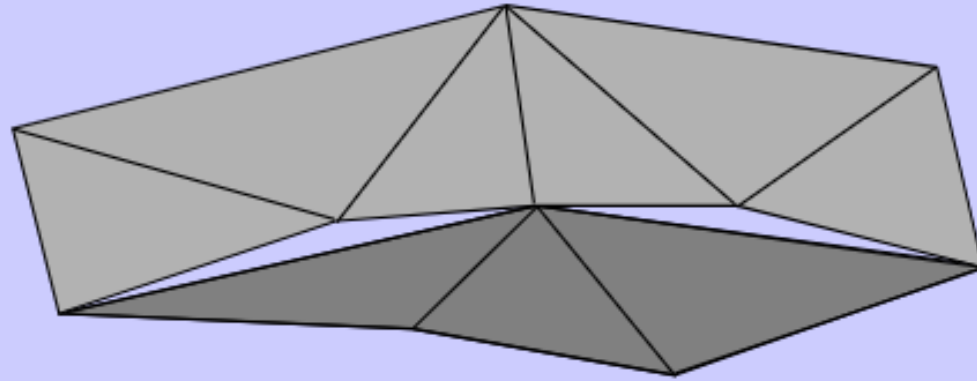
Perfect Resolution



STL File Problems

- Several problems plague STL files and they are due to the very nature of STL files.
- Reason for problems is that .stl file **contain no topological data**.
- Tessellation algorithms used by CAD vendors as they tend to create polygonal approximation models which exhibit the following types of errors:
 - ✓ Gaps (cracks, holes, punctures) that is, missing facets.
 - ✓ Degenerate facets (where all its edges are collinear).
 - ✓ Overlapping facets.
 - ✓ Non-manifold topology conditions.

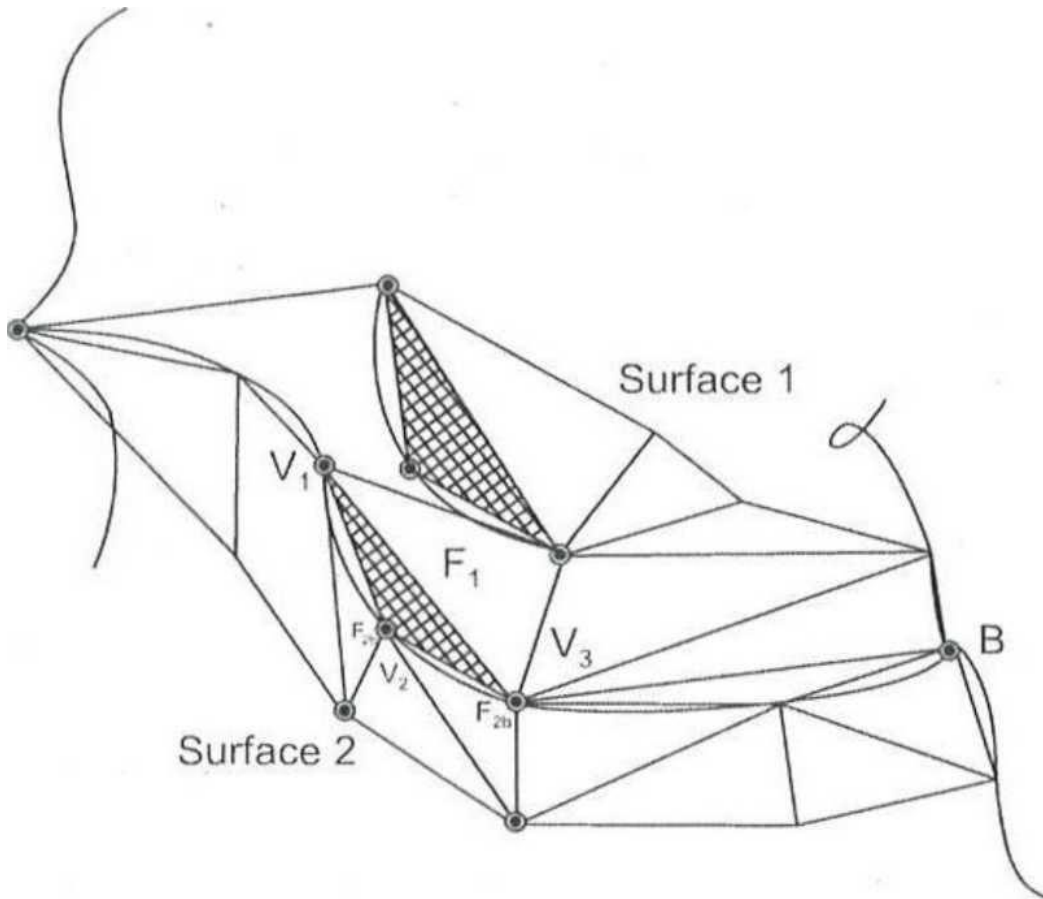
Shell Punctures



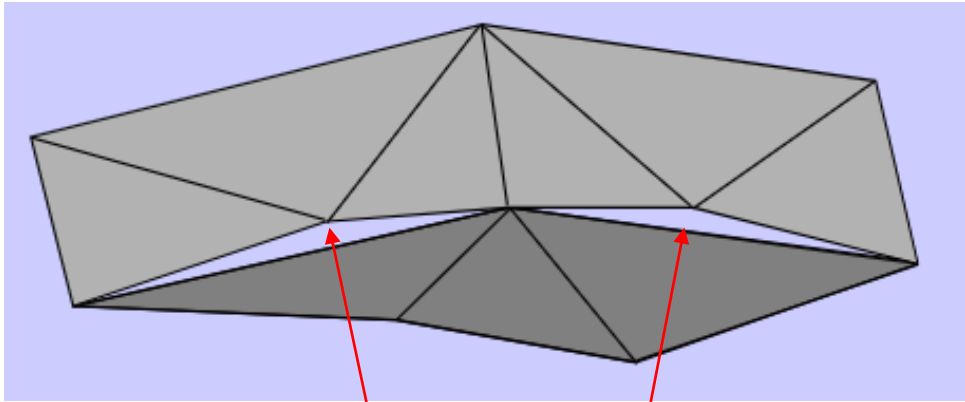
- Holes in part boundary often caused by uneven tessellation of surfaces
- Can cause scan line problems

Missing Facets or Gaps

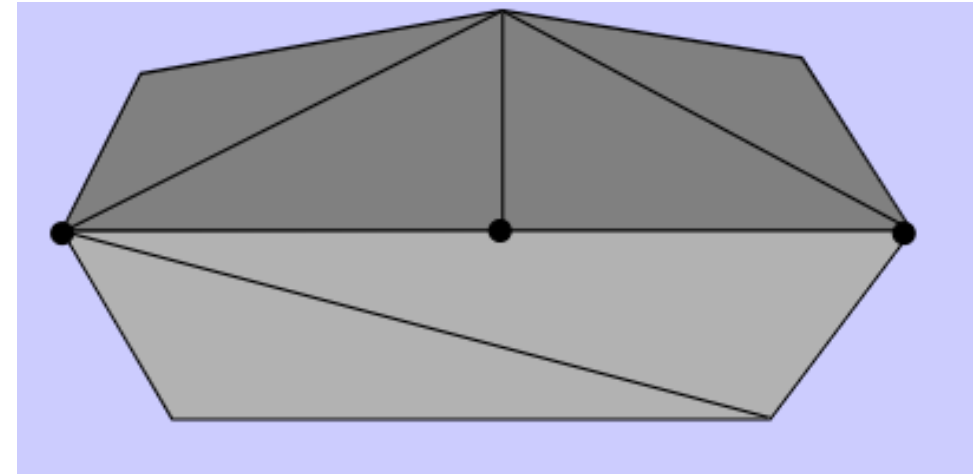
same as previous one



- Tessellation of surfaces with large curvature can result in errors at the intersections between such surfaces, leaving gaps or holes along edges of the part model.



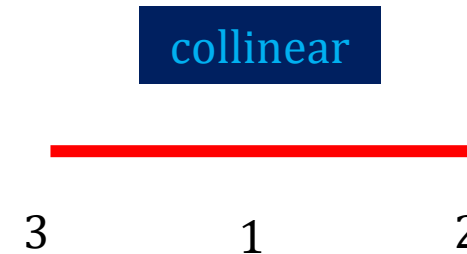
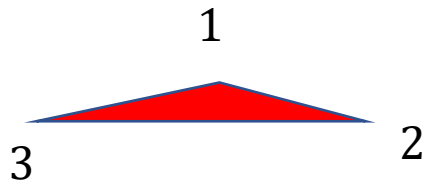
To avoid shell punctures



Degenerate faces
(Collinear vertices)

Degenerate Facets

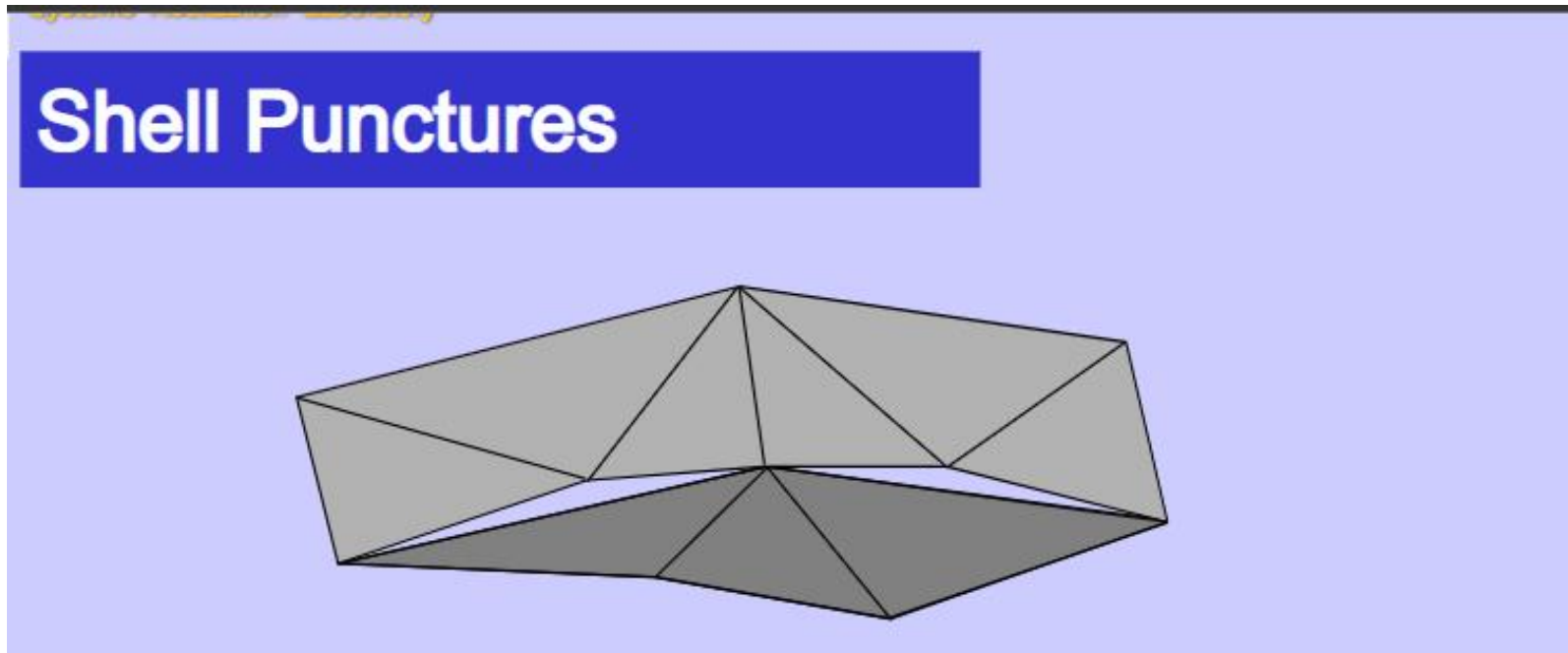
- A geometrical degeneracy of a facet occurs when all of the **facets' edges** are **collinear** even though all its **vertices are distinct**.
- This might be caused by stitching algorithms that attempt to avoid shell punctures
- The resulting facets generated eliminate the **shell punctures**.



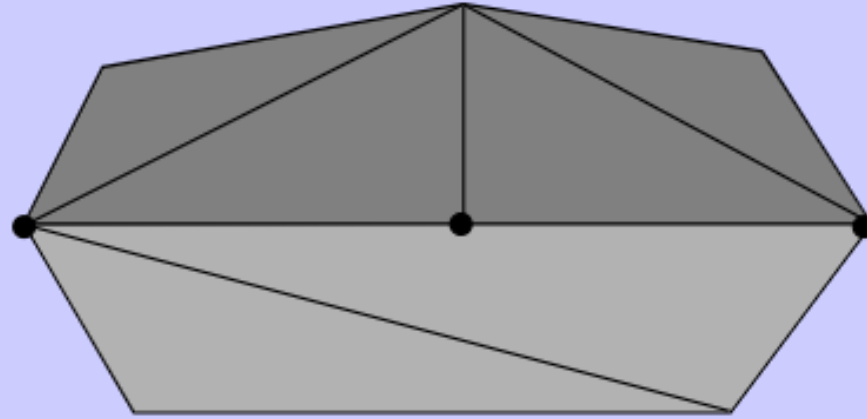
However, this is done at the expense of adding a **degenerate facet**.

While degenerate facets **do not contain valid surface normals**, they do represent implicit topological information on how two surfaces mated.

This important information is consequently stored prior to discarding the degenerate facet.



Degenerate Facets - Geometric



- 3 distinct but collinear vertices
- Added by some stitching algorithms in order to avoid shell punctures.
- Undefined surface normal.
- Cannot just delete facet because it is important topologically.

Degenerate Facets

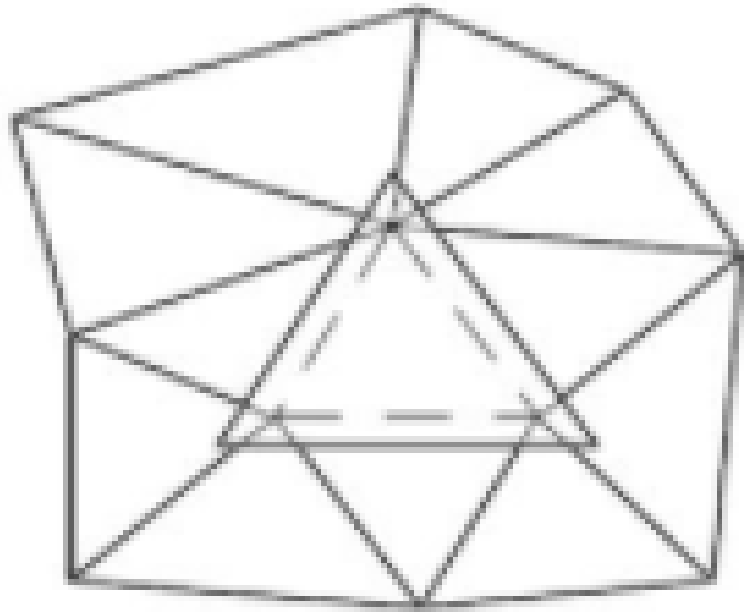


Before



After

Overlapping Facets



- Overlapping facets may be generated due to numerical **roundoff errors** occurring during tessellation.
- The vertices are represented in 3D spaces as **floating point numbers** instead of **integers**. Thus the numerical roundoff can cause facets to overlap if tolerances are set too liberally.

NON-MANIFOLD CONDITIONS

- There are three types of non-manifold conditions, namely:
 - (1) A non-manifold edge.
 - (2) A non-manifold point.
 - (3) A non-manifold face.
- These may be generated because tessellation of the fine features are susceptible to round-off errors.

- Here, the non-manifold edge is actually shared by four different facets.
- A valid model would be one whose facets have only an adjacent facet each, that is, one edge is shared by two facets only.
- Hence the non-manifold edges must be resolved such that each facet has only one neighboring facet

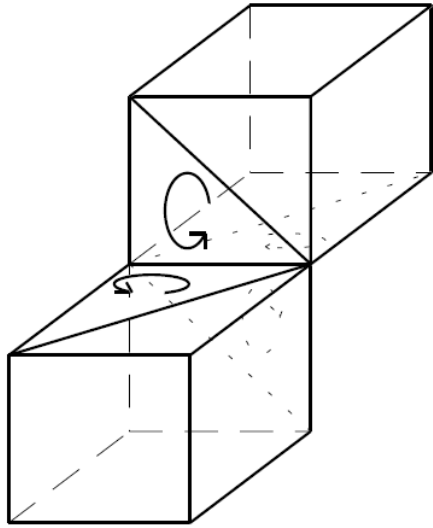


Figure 6.6(a): A non-manifold edge whereby two imaginary minute cubes share a common edge

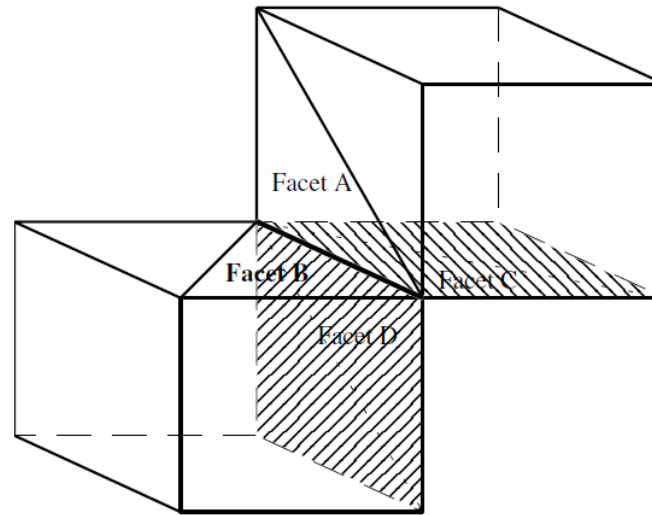
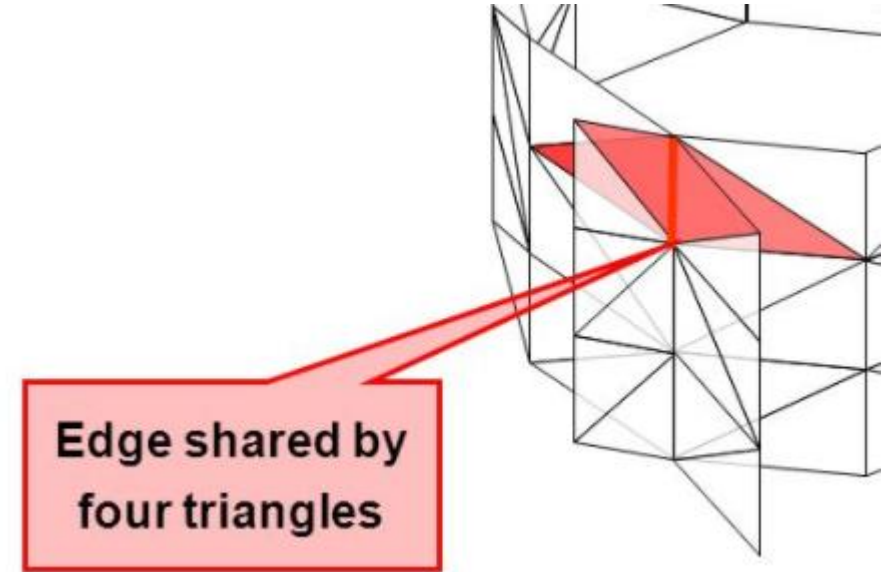


Figure 6.6(b): A non-manifold edge whereby four facets share a common edge after tessellation



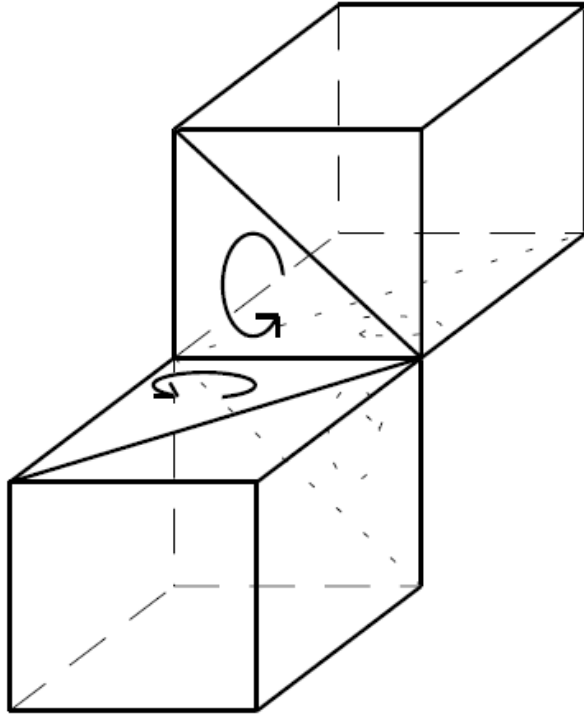


Figure 6.6(a): A non-manifold edge whereby two imaginary minute cubes share a common edge

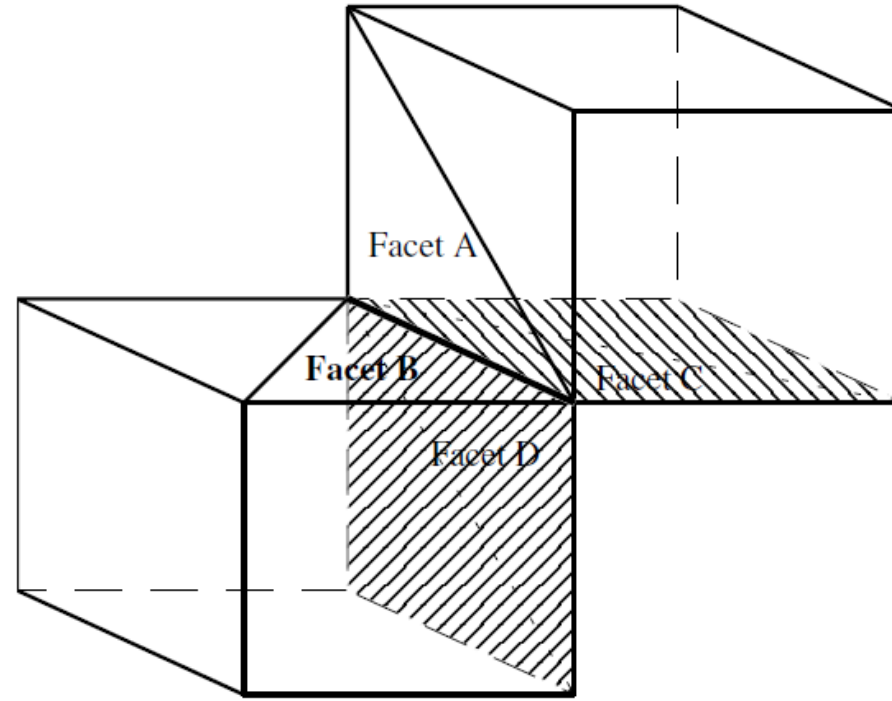


Figure 6.6(b): A non-manifold edge whereby four facets share a common edge after tessellation

Non-manifold point & Non-manifold face

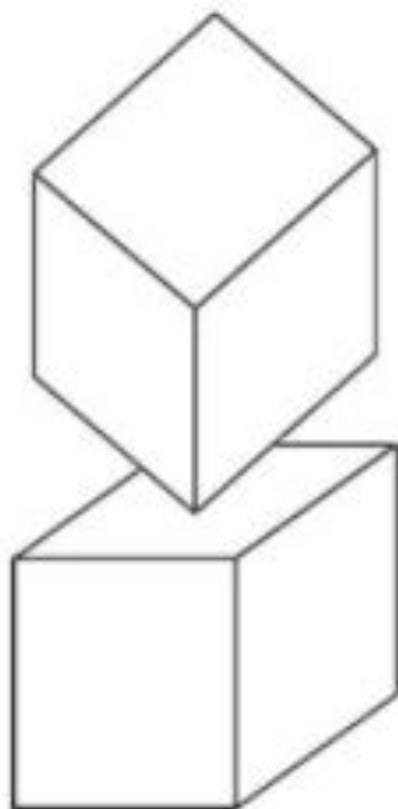


Figure 6.6(c): Non-manifold point

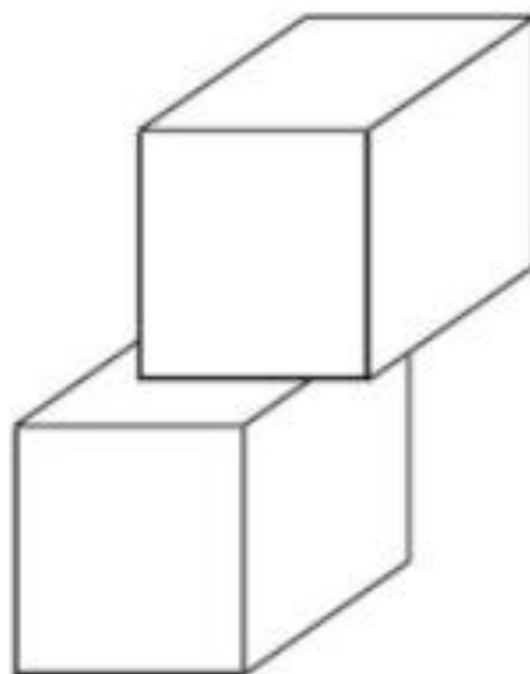
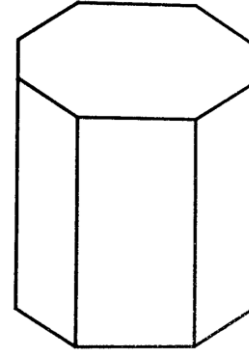


Figure 6.6(d): Non-manifold face

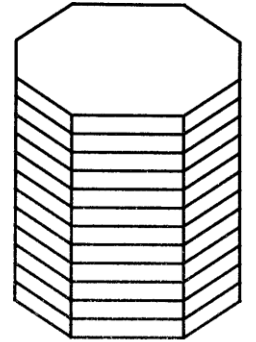
If **Valid Model** is Available:

A tessellated model is said to be valid if there are **no missing facets, degenerate facets, overlapping facets, or any other abnormalities**.

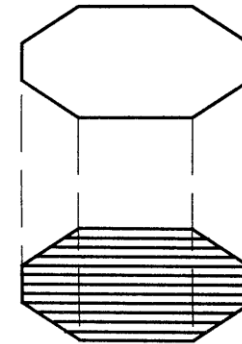
- When a valid tessellated model is used as an input, it will first be sliced into **2D layers**.
- Each layer would then be converted into **unidirectional scan lines** for the laser or other RP techniques to commence building the model.
- The scan lines would act as on/off points for the laser beam controller so that the part model can be built accordingly without any problems.



A valid
3D model



A 3D
model sliced into
2D planar layers



Conversion
of 2D layers into
1D scan lines

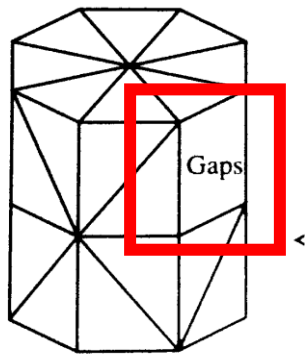
If tessellated model is **invalid**:

A solid model is tessellated non-robustly and results in a gap.

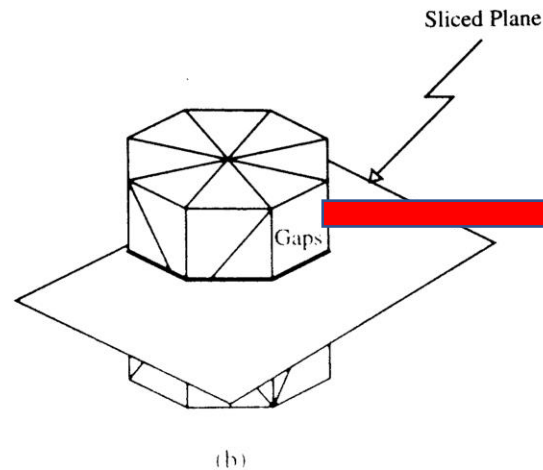
If this error is not corrected and the model is subsequently sliced, in preparation for it to be built layer by layer, the missing facet in the geometrical model would cause the system to have **no pre-defined stopping** boundary on the particular slice.

Thus building process would continue right to the physical limit of the RP machine, creating a stray physical solid line and ruining the part being produced.

Therefore, it is of paramount importance that the model be “repaired” before it is sent for building.

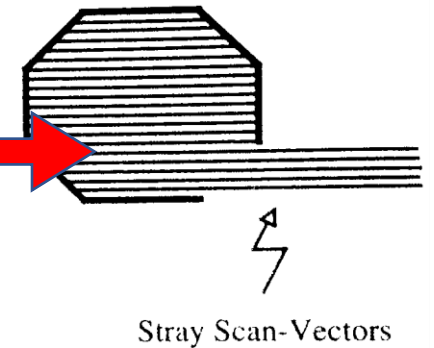


An invalid
tessellated model



An invalid model
being sliced

TOP VIEW



no pre-defined stopping